

NAME

codata - library for fundamental physical constants

LIBRARY

codata (-libcodata, -lcodata)

SYNOPSIS

```
use codata
include "codata.h"
import pycodata
```

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA (<https://www.nist.gov/programs-projects/codata-values-fundamental-physical-constants>). A C API allows usage from C, or can be used as a basis for other wrappers. A python wrapper allows easy usage from Python.

The latest *codata* constants (2022) <https://pml.nist.gov/cuu/Constants> were integrated in stdlib <https://github.com/fortran-lang/stdlib/releases/tag/> since version 0.7.0. The constants are implemented as derived type which carries the name, the value, the uncertainty and the unit. This library is complementary to the constants defined in the stdlib by providing older values for the constants. The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix in their name.

All *codata* (physical) constants are defined as a derived type `codata_constant_type`. All the *codata* constants are provided as double precision reals. The names are quite long and can be aliased with shorter names. The derived type `codata_constant_type` defines 4 members and 2 procedures.

```
type :: codata_constant_type
  character(len=64) :: name
  real(dp) :: value
  real(dp) :: uncertainty
  character(len=32) :: unit
contains
  procedure :: print
  procedure :: to_real_sp
  procedure :: to_real_dp
  generic :: to_real => to_real_sp, to_real_dp
end type codata_constant_type
```

A module level interface `to_real` is available for getting the constant value or uncertainty of a constant.

o `to_real(self, mold, uncertainty) result(r)`

Get the constant value or uncertainty. Warning: Some constants cannot be converted to single precision sp reals due to the value of the exponents.

o `class(codata_constant_type), intent(in) :: self`

Codata constant

o `real(sp), intent(in) :: mold`

Should be of type real single or double precision

o `logical, intent(in), optional :: uncertainty`

Set to true if the uncertainty is required. Default to .false..

The C API exposes a structure `codata_constant_type` that defines the same members as in Fortran. `to_real` is not exposed in the C API.

```
type, bind(C) :: capi_constant_type
  character(kind=c_char) :: name(65)
  real(c_double) :: value
```

```

        real(c_double) :: uncertainty
        character(kind=c_char) :: unit(33)
    end type capi_constant_type

    typedef struct codata_constant_type{
        char name[65];
        double value;
        double uncertainty;
        char unit[33];
    }cct;

```

The Python wrapper encapsulates the members in a dictionary with the keys being name, value, uncertainty and unit. `to_real` is not exposed in the Python wrapper.

NOTES

To **use** *codata* within your **fpm**(1) (<https://github.com/fortran-lang/fpm>) project, add the following lines to your file:

```

[dependencies]
codata = { git="https://github.com/MilanSkocic/codata.git" }

```

EXAMPLE

Example in Fortran

```

program example_in_f
use codata
implicit none
print '(A)', '##### EXAMPLE IN FORTRAN #####'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ", SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ", MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ", MOLAR_MASS_CONSTANT_2010%value
end program

```

Example in C

```

#include <stdio.h>
#include "codata.h"
int main(void){
printf("##### EXAMPLE IN C #####0);
printf("%s0,"# VERSION");
printf("version = %s0, codata_version());
printf("%s0,"# CONSTANTS");
printf("c = %f0, SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s0,"# UNCERTAINTY");
printf("u(c) = %f0, SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s0,"# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f0, MOLAR_MASS_CONSTANT.value);

```

```
printf("Mu_2018 = %23.16f0, MOLAR_MASS_CONSTANT_2018.value);  
printf("Mu_2014 = %23.16f0, MOLAR_MASS_CONSTANT_2014.value);  
printf("Mu_2010 = %23.16f0, MOLAR_MASS_CONSTANT_2010.value);  
return 0;  
}
```

Example in Python

```
import sys  
sys.path.insert(0, "../py/src/")  
import pycodata  
print("##### EXAMPLE IN PYTHON #####")  
print("# VERSION")  
print(f"version = {pycodata.__version__}")  
print("# Constants")  
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])  
print("# UNCERTAINTY")  
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])  
print("# OLDER VALUES")  
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])  
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])  
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])  
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

SEE ALSO

gsl(3), **codata(1)**, **fpm(1)**